
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Touch Display
Manufacturer: CTOUCH
Firmware Version: N/A
Model(s): Laser Sky, Laser Nova

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_1_0_0	10/11/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '255' and 'Broadcast'. DeviceID 138 and 168 are not supported on the device. When DeviceID is set to 'Broadcast', status is not available.
Example: `InterfaceName.DeviceID = '1'`
- Only Serial supports status

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Laser Sky')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Laser Sky')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 5000, Model='Laser Sky')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'HDMI 1' 'HDMI 4 / RK Android' ¹ 'Inside PC'	Value 'HDMI 2' 'DisplayPort' 'Home / COS'	Value 'HDMI 3' 'VGA'
# Input example InterfaceName.Set('Input', 'HDMI 1')			
Command MenuNavigation	Value 'Menu' 'Left' 'Exit'	Value 'Up' 'Right'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command PictureMode	Value 'Dynamic' 'User'	Value 'Standard' 'Writing'	Value 'Soft'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

Command Screenshot	Value None
# Screenshot example InterfaceName.Set('Screenshot', None)	
Command Volume	Value 0 to 100 in steps of 1
# Volume example InterfaceName.Set('Volume', 100)	

¹ HDMI 4 is supported by Laser Sky when the Wi-Fi module is inserted and RK Android is supported by Laser Nova.

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'HDMI 1'	Value 'HDMI 2'	Value 'HDMI 3'
	'HDMI 4 / RK Android' ¹	'DisplayPort'	'VGA'
	'Inside PC'	'Home / COS'	
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command PictureMode	Value 'Dynamic'	Value 'Standard'	Value 'Soft'

Global Scripter Module Communication Sheet

	'User'	'Writing'
<pre># PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)</pre>		
Command	Value	Value
Power	'On'	'Off'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command	Value	
Volume	0 to 100 in steps of 1	
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>		

¹ HDMI 4 is supported by Laser Sky when the Wi-Fi module is inserted and RK Android is supported by Laser Nova.

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Freeze_TCP	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'HDMI 1' 'HDMI 4 / RK Android' ¹ 'Inside PC'	Value 'HDMI 2' 'DisplayPort' 'Home / COS'	Value 'HDMI 3' 'VGA'
# Input example InterfaceName.Set('Input', 'HDMI 1')			
Command MenuNavigation	Value 'Menu' 'Left' 'Exit'	Value 'Up' 'Right'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command PictureMode	Value 'Dynamic' 'User'	Value 'Standard' 'Writing'	Value 'Soft'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command PowerOff	Value None		
# PowerOff example InterfaceName.Set('PowerOff', None)			

Command Screenshot	Value None
# Screenshot example InterfaceName.Set('Screenshot', None)	
Command Volume	Value 0 to 100 in steps of 1
# Volume example InterfaceName.Set('Volume', 100)	

¹ HDMI 4 is supported by Laser Sky when the Wi-Fi module is inserted and RK Android is supported by Laser Nova.

Status Available (Ethernet)

N/A

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

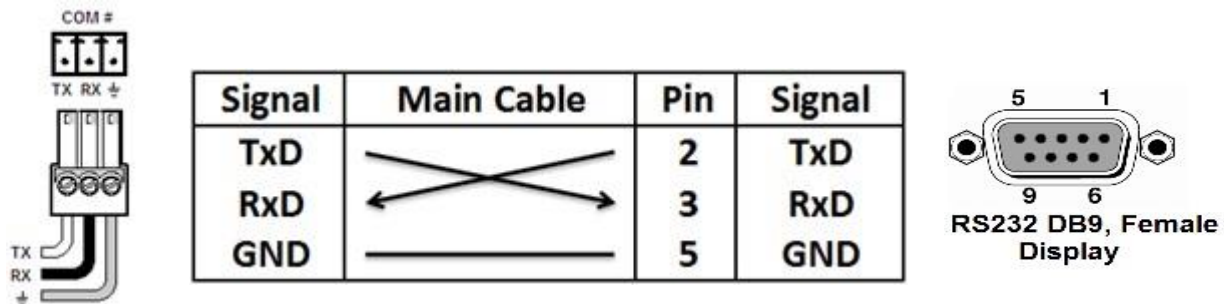
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	5000
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Audio Mute Off	\xA9\x14\x01\x00\x8A
Audio Mute On	\xA9\x14\x01\x01\x8A
Executive Mode Off	\xA9\x17\x01\x00\x8A
Executive Mode On	\xA9\x17\x01\x01\x8A
Freeze Off	\xA9\x1B\x01\x00\x8A
Freeze On	\xA9\x1B\x01\x01\x8A
Input DisplayPort	\xA9\x15\x01\x07\x8A
Input HDMI 1	\xA9\x15\x01\x05\x8A
Input HDMI 2	\xA9\x15\x01\x06\x8A
Input HDMI 3	\xA9\x15\x01\x08\x8A
Input HDMI 4 / RK Android	\xA9\x15\x01\x0D\x8A
Input Home / COS	\xA9\x15\x01\x0C\x8A
Input Inside PC	\xA9\x15\x01\x09\x8A
Input VGA	\xA9\x15\x01\x14\x8A
Menu Navigation Down	\xA9\x16\x01\xD8\x8A
Menu Navigation Enter	\xA9\x16\x01\x9B\x8A
Menu Navigation Exit	\xA9\x16\x01\xD4\x8A
Menu Navigation Left	\xA9\x16\x01\x97\x8A
Menu Navigation Menu	\xA9\x16\x01\x84\x8A
Menu Navigation Right	\xA9\x16\x01\x9F\x8A
Menu Navigation Up	\xA9\x16\x01\x92\x8A
Picture Mode Dynamic	\xA9\x18\x01\x00\x8A
Picture Mode Soft	\xA9\x18\x01\x02\x8A
Picture Mode Standard	\xA9\x18\x01\x01\x8A
Picture Mode User	\xA9\x18\x01\x03\x8A
Picture Mode Writing	\xA9\x18\x01\x06\x8A
Power Off	\xA9\x11\x01\x00\x8A
Power On	\xA9\x11\x01\x01\x8A
Screenshot None	\xA9\x16\x01b\x8A
Volume 0	\xA9\x13\x01\x00\x8A
Volume 100	\xA9\x13\x01d\x8A

Appendix B. Update Commands (Serial)

Audio Mute	\xA9\x14\x01\xAA\x8A
Executive Mode	\xA9\x17\x01\xAA\x8A
Freeze	\xA9\x1B\x01\xAA\x8A
Input	\xA9\x15\x01\xAA\x8A
Picture Mode	\xA9\x18\x01\xAA\x8A
Power	\xA9\x11\x01\xAA\x8A
Volume	\xA9\x13\x01\xAA\x8A

Appendix C. Set Commands (Ethernet)

Audio Mute Off	\xA9\x14\x01\x00\x8A
Audio Mute On	\xA9\x14\x01\x01\x8A
Executive Mode Off	\xA9\x17\x01\x00\x8A
Executive Mode On	\xA9\x17\x01\x01\x8A
Freeze Off	\xA9\x1B\x01\x00\x8A
Freeze On	\xA9\x1B\x01\x01\x8A
Input DisplayPort	\xA9\x15\x01\x07\x8A
Input HDMI 1	\xA9\x15\x01\x05\x8A
Input HDMI 2	\xA9\x15\x01\x06\x8A
Input HDMI 3	\xA9\x15\x01\x08\x8A
Input HDMI 4 / RK Android	\xA9\x15\x01\x0D\x8A
Input Home / COS	\xA9\x15\x01\x0C\x8A
Input Inside PC	\xA9\x15\x01\x09\x8A
Input VGA	\xA9\x15\x01\x14\x8A
Menu Navigation Down	\xA9\x16\x01\xD8\x8A
Menu Navigation Enter	\xA9\x16\x01\x9B\x8A
Menu Navigation Exit	\xA9\x16\x01\xD4\x8A
Menu Navigation Left	\xA9\x16\x01\x97\x8A
Menu Navigation Menu	\xA9\x16\x01\x84\x8A
Menu Navigation Right	\xA9\x16\x01\x9F\x8A
Menu Navigation Up	\xA9\x16\x01\x92\x8A
Picture Mode Dynamic	\xA9\x18\x01\x00\x8A
Picture Mode Soft	\xA9\x18\x01\x02\x8A
Picture Mode Standard	\xA9\x18\x01\x01\x8A
Picture Mode User	\xA9\x18\x01\x03\x8A
Picture Mode Writing	\xA9\x18\x01\x06\x8A
Power Off None	\xA9\x11\x01\x00\x8A
Screenshot None	\xA9\x16\x01b\x8A
Volume 0	\xA9\x13\x01\x00\x8A
Volume 100	\xA9\x13\x01d\x8A